

RESEARCH ARTICLE

Additive Manufacturing of an Anthropomorphic Hand using Force Resistive Sensors for Classification of Four Distinct Objects

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ABSTRACT: Anthropomorphic robotic hands have become increasingly popular for applications in recent decades. They are used in the development of prosthetics, surgical robots, and teleoperation. However, existing robotic hands are currently limited by grip force sensitivity and expense. A clinically prescribed and commercially available (CPCA) prosthetic can cost upwards of \$20,000 USD. This study used multimaterial 3D printing and low-cost film pressure sensors to develop a high-performance, low-cost anthropomorphic hand. For 3D printing, PLA (Polylactic acid) was used to create the fingers and arm, and TPU (Thermoplastic Polyurethane) was used to create the joints and palm. Integrated force resistive sensors were implemented to detect pressure changes while grabbing objects. Training data was collected by grabbing four uniquely shaped objects, then a random forest machine learning algorithm was used to identify the objects based on sensor inputs. The model sustained an accuracy of 98.1% from 4500 rows of training data and 1000 rows of testing data. An analysis of pressure values and grip trends supports the hypothesis that additive manufacturing can be combined with force resistive sensors to create a low-cost bioinspired hand with object detection capabilities. Accurate results in object detection may be expanded on with potential applications in the advancement of adaptive grasp, soft robotics, smart prosthetics, AI, human skin biomimicry, and the consideration of material properties in computer vision.

KEYWORDS: additive manufacturing, anthropomorphic hand, force resistive sensors, object detection, random forest machine learning

INTRODUCTION

The human hand is considered a highly dextrous manipulator, with the ability to grasp or hold objects ranging in size, texture, and geometry [7]. Giving human characteristics and appearances, like the shape, structure, and ability of the human hand, to inanimate objects is called anthropomorphism [18]. Anthropomorphic robotic hands are used in applications such as prosthetics, surgery, and teleoperation. In these fields, the ability of robotic hands to explore and modify their environment to a higher degree has grown in demand [8]. Including the integration of human hand characteristics into the design of robotic hands, the dexterity and precision of grabbing can be improved. Current uses of anthropomorphic robotic hands include open-source 3D printed hands like the HACKberry Hand, the HANDi Hand, and the BEAR PAW [17]; prosthetics; robotic hand and arm structures for precise and minimally invasive surgeries [20]; and teleoperation, where robotic hands and arms are controlled by humans [3]. High quality CPCA prosthetic devices can cost more than \$20,000 USD [17]. Meanwhile, low-cost 3D printed hands face significant limitations including inadequate grip strength, insufficient friction, and design factors that prevent the effective reorientation of contact forces [17]. Choosing a more cost-effective actuator or controller oftentimes decreases hand functionality. For balancing cost with quality of product, additive manufacturing is a viable low-cost alternative to traditional methods like subtractive or formative manufacturing.

Additive manufacturing has rapidly grown into a widely used technology. The process involves the physical creation of a digital three-dimensional model created by Computer-Aided

Design (CAD) through the addition of material layer by layer [16]. There are many types of additive manufacturing including Stereolithography (SLA) and Binder Jetting, but the most common type of additive manufacturing, which was used for our anthropomorphic hand, is Fused Deposition Modeling (FDM), also known as 3D printing. FDM creates objects by feeding material in the form of filament into an extruder, then heating the filament and extruding it onto bed layers [19].

The technology of 3D printing has advanced rapidly in accessibility, precision, and viability. 3D printing is both cost effective and fast, allowing for rapid prototyping [10]. It also allows the creation of more complex shapes without additional cost, increasing design customization capabilities [1]. These advantages have led 3D printing to be applied to a variety of industries. It is used to create lightweight and affordable parts in the aerospace and automotive industries or to design easily customized parts for electronics or to more efficiently print environmentally friendly buildings. 3D printing is also used to create skin, tissues, bone, organs, and cartilage for the healthcare and medical industries. It has even been implemented in the fashion and food industries, allowing for 3D printed shoes, jewelry, and clothes as well as customizable chocolate, pasta, and pizza [16].

For the designing and manufacturing of a robotic hand, 3D printing is affordable and efficient for making many design iterations. Since it also allows for the creation of detailed designs, it is well suited for the creation of the complex structure of an anthropomorphic hand. The joints in the fingers are a

fundamental part of the hand, and 3D printing allows for the creation of soft joints that can mimic natural human joints [14]. In addition to this, 3D printing also offers a variety of materials that can range from commercial levels, such as Polylactic acid (PLA) and Thermoplastic Polyurethane (TPU) to industrial-grade metals, ceramics, polymers, and composites.

Robotic hands are inspired by human hands because of their efficiency and versatility. The unique structure and movement of the human hand, with bones, joints, muscles, tendons, and nerves, allows for precise function. 3D printed fingers contain flexible soft joints, and strings and motors mimic muscle and tendon movement [9,15]. Robotic hands can grasp objects of different shapes, weights, and sizes without damage. This is useful in many fields including healthcare, manufacturing, and service [4,5]. Adapting components of the human hand also helps robotic hands have more natural and efficient control

systems [9]. This hand contains all distal interphalangeal, proximal interphalangeal, and metacarpophalangeal joints. The size and shape of this robotic hand was scaled from the metacarpal dimensions of our team member Christopher Huang's hand. Bioinspiration in joints and palm flexibility allows for the robotic hand to mimic the dexterity of a human hand's geometry. It can complete complex tasks by copying the flexibility and adaptability of human hands. The use of sensors can aid robotic hands in reaction speed and accuracy by detecting contact patterns. Being able to sense pressure mimics the function of mechanoreceptors in the skin [6,21]. Additionally, we have considered artificial skin with fingerprint ridges and skin creases modelled after existing applications of tactile sensing electronic skin [11]. The texture will help the robotic hand grip with less strength and more precision [2].

Figure 1

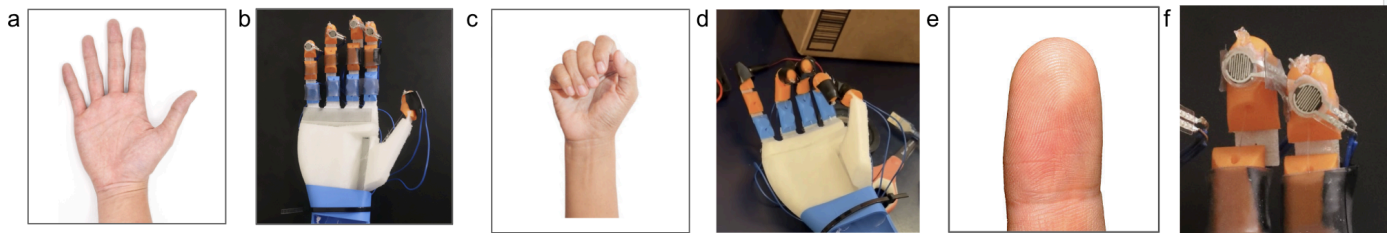


Figure 1: The bioinspiration in shape, structure, and size from a human hand, as well as the similarities in hand function and movement. “a” and “b” show the likeness between the size and shape of a human hand and the robotic hand, “c” and “d” show the similarities in hand movement, and “e” and “f” show the correspondence in hand function.

To begin the creation of the robotic hand, inspired by a real human hand as shown in Figure 1, the parts were designed through CAD and then 3D printed. 3D printing is a popular method for simple prototyping, testing, and improving, due to its quick manufacturing time and inexpensive costs. Different materials were used for different functions and areas of the hand. The combination of rigid and flexible materials creates a hand that is durable and adaptable. The hand has grabbing strength similar to a normal hand and can hold objects of various shapes. This finger claw system contains force resistive sensors on the fingertips that produce feedback on pressure readings from objects (see Figure 1f). An electronic system in and on the forearm of the hand contains tactile sensors for detection and servo motors for grasping. The robotic hand can grab different objects and collect data via the sensors (see Figure 1b and 1d). We used this data to train a random forest machine learning model for object identification. This model is highly effective for classification tasks where it makes decisions based on observed previous patterns. The random forest machine learning model reduces error through decision trees, works well with smaller data sets, and is able to handle sensor fluctuations. Random forest models are accurate and easily trainable, allowing for applications that can use object detection to specifically read grip strength or perform further instructions. This anthropomorphic hand costs about \$103.07 USD to produce, making it viable for low-cost tactile sensing, and the object detection model expands the design for further applications in agriculture, cooking, research, and computer vision.

MATERIALS AND METHODS

All components were created using the computer-aided design (CAD) softwares Onshape and Fusion 360. The models were then exported as a stereolithography (STL) file and imported into the slicing software Bambu Studio. Fused deposition modeling (FDM) parts were printed on a Bambu Lab P1S (build volume: 256 x 256 x 256 mm) and a Bambu Lab A1 mini (build volume: 180 x 180 x 180 mm). A 0.4 mm nozzle was used for all parts and PLA components were printed with a nozzle temperature of 220 °C, while TPU components were printed at 240 °C. The overall size of the assembled model was 466 mm in length and 149 mm in width. Individual PLA fingers ranged from 29mm in length for the thumb to 97mm for the middle finger. The TPU finger joints measured 8mm in length, while the TPU palm measures at 107mm vertically. The wrist/forearm piece measures at 239mm in length. The wheels vary in diameter from 32mm for the thumb to 35mm for the pointer finger. Print settings were optimized based on the function of each component. PLA fingers and wheel components were printed with a grid pattern and 15% infill. TPU joints were also printed with a 15% grid infill pattern. The TPU palm was printed with a 6% gyroid infill pattern. The wrist and forearm were printed with an infill of 8%. After FDM printing, all parts went through post processing, including removing supports and surface cleaning as needed. The fingertip caps were printed with EL 400 Resin on a Phrozen Sonic Mini 8K stereolithography (SLA) printer. The files were prepared for printing using the slicing software Chitubox. After printing, the parts were removed off the build plate, washed in a cleaning solution for 10 minutes and cured with UV light for between 5 and 10 minutes.

Figure 2

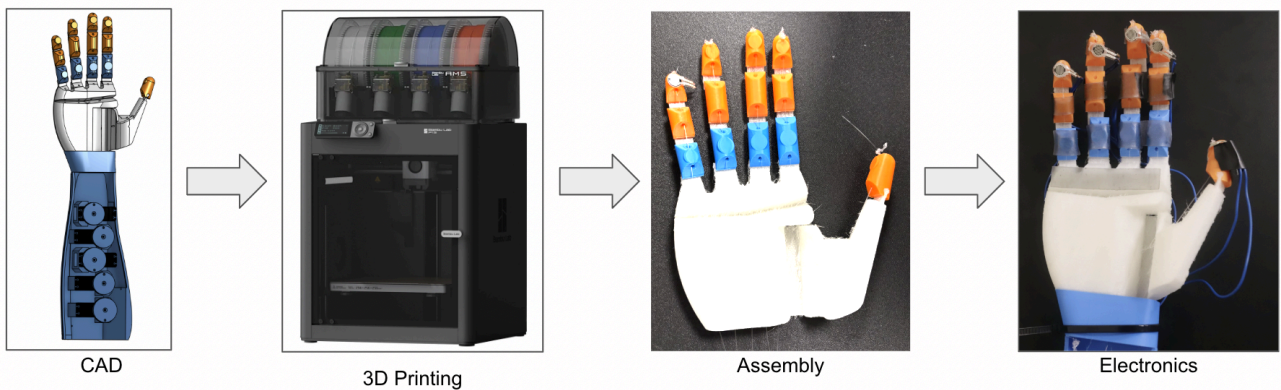


Figure 2: The flow chart of the design and assembly of the anthropomorphic hand. First, the whole hand was modeled in CAD, then 3D printed on a Bambu Lab P1S or A1 mini. Then it was assembled, first by the fingers, which were then added to the palm and strung with fishing line so the fingers would contract. Then it was attached to the wrist and forearm, and the electronics were soldered and attached.

As shown in Figure 3, the hand is controlled by an Arduino Uno, which can be powered by a nine-volt battery or through the connection to a computer. A 6V Tenenergy battery pack powers a pin board, which has five Miuzei servos and five USDRWAM pressure sensors connected with unleaded solder. The pinboard was made by first soldering in a positive power rail and a negative power rail from the Tenenergy battery pack, then by soldering the positive pressure sensor wires into the board, then to the power rail. The negative end of the pressure sensor was then soldered into the board in the same vertical row as the communication wire and the 10 kilo-ohm resistor, which were then all connected with solder to ensure a strong connection. The communication wire, which reads the voltage drop across the

sensor and outputs it as a resistance value, was then plugged into the Analogue ports of the Arduino Uno. Finally, the other end of the resistor was soldered to the negative power rail of the board. For the servos, the positive power wire was soldered first into the board, and then to the power rail, with the communications wire, which delivers instructions to the motor, skipping the pin board entirely and just plugging into the PWM (Pulse Width Modulation) ports on the Arduino Uno, designated with a ~ in front of the number. Then the negative wire was soldered to the pin board and then to the negative power rail.

Figure 3

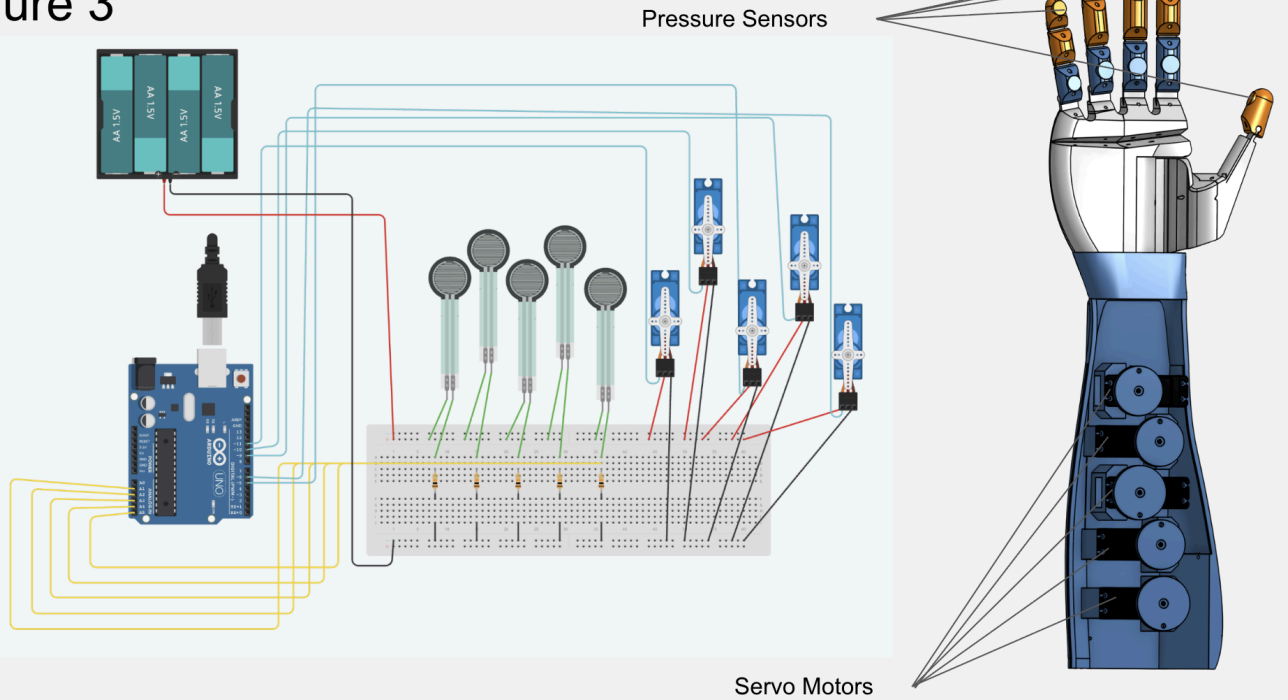


Figure 3: The wiring diagram and setup of the robotic hand, with the 6V battery pack powering the pinboard with the pressure sensors and servos, as well as where the pressure sensors and servos connect on the hand.

The robotic hand was tested through force resistive sensor data after grabbing four different objects. A chip bag weighing 31.7641 grams, a Rice Krispy treat weighing 22.1126 grams, a Pringles container weighing 25.5187 grams, and a Sharpie box weighing 15.9577 grams. In testing, the hand was held against the hand mimicking natural actions while maximizing sensor surface area in the grip. After running code to grab the object, we let go so only the robot hand is holding it and begin data collection from the sensors. The force resistive sensors are FSR/RP-C type sensors, which correlate decreased resistance with increased force. The arduino maps the resistance from the FSR to a value through its ADC using the voltage divider formula

$V_{out} = V_{cc} * (R_{fixed} / (R_{FSR} + R_{fixed}))$, where V_{cc} is the supply voltage of 5V, R_{fixed} is the 10 kilohm resistor in the circuit, and the R_{FSR} is the resistance from the force-resistive sensor. This equation changes the output values of resistance to be increasing as the grip force increases. The data was saved in Excel as one set of inputs from every finger every 0.1 seconds. The grip was the same each test, and multiple tests were conducted for a total of around 5500 lines of data. The data measures resistance which is positively correlated with the force of grabbing the object.

Figure 4

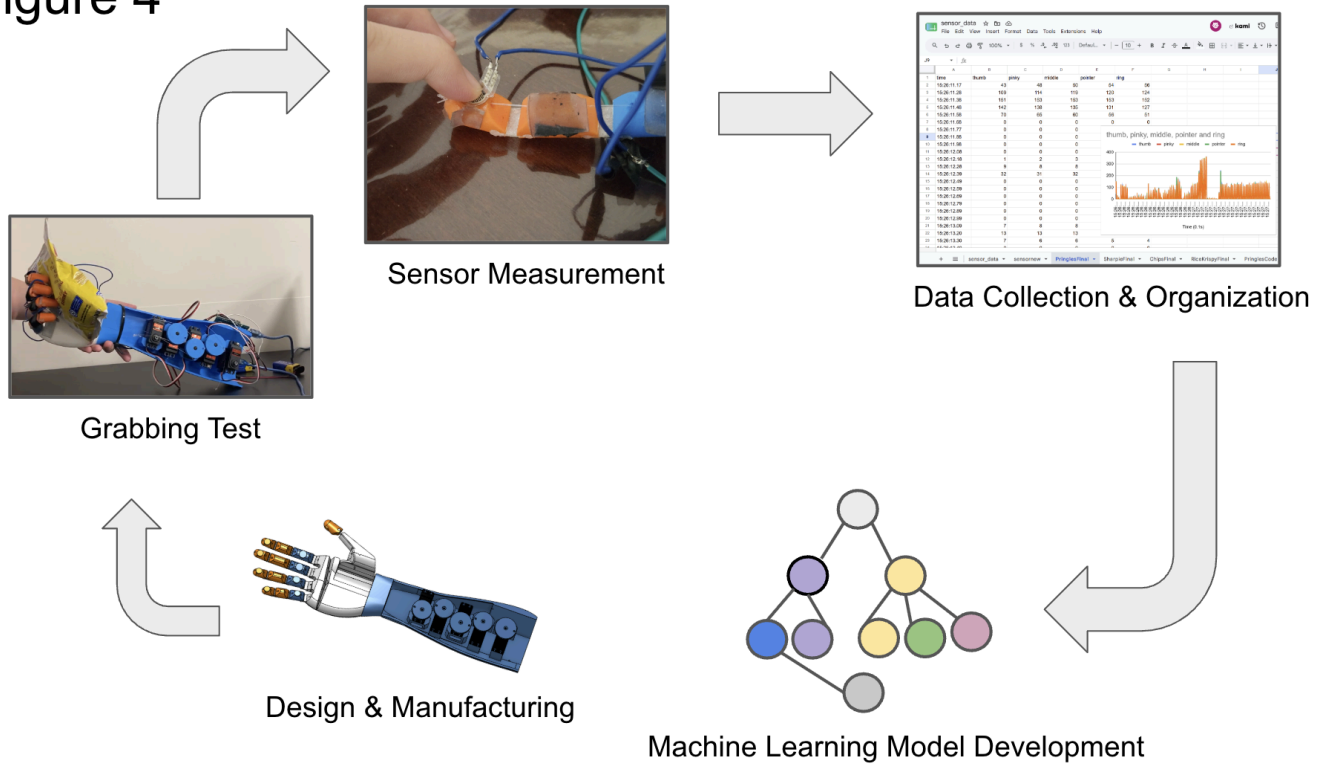


Figure 4: The flow chart that led to the training and development of the Random Forest Machine Learning Model. We started with the design and manufacturing of the anthropomorphic hand, which led to a grabbing test to ensure we could properly pick up the objects. The success with the grabbing test led to grabbing the objects with the pressure sensors running, which allowed us to collect and organize the data readings, and put them into a machine learning model to train it for object detection.

RESULTS AND DISCUSSION

Figure 5

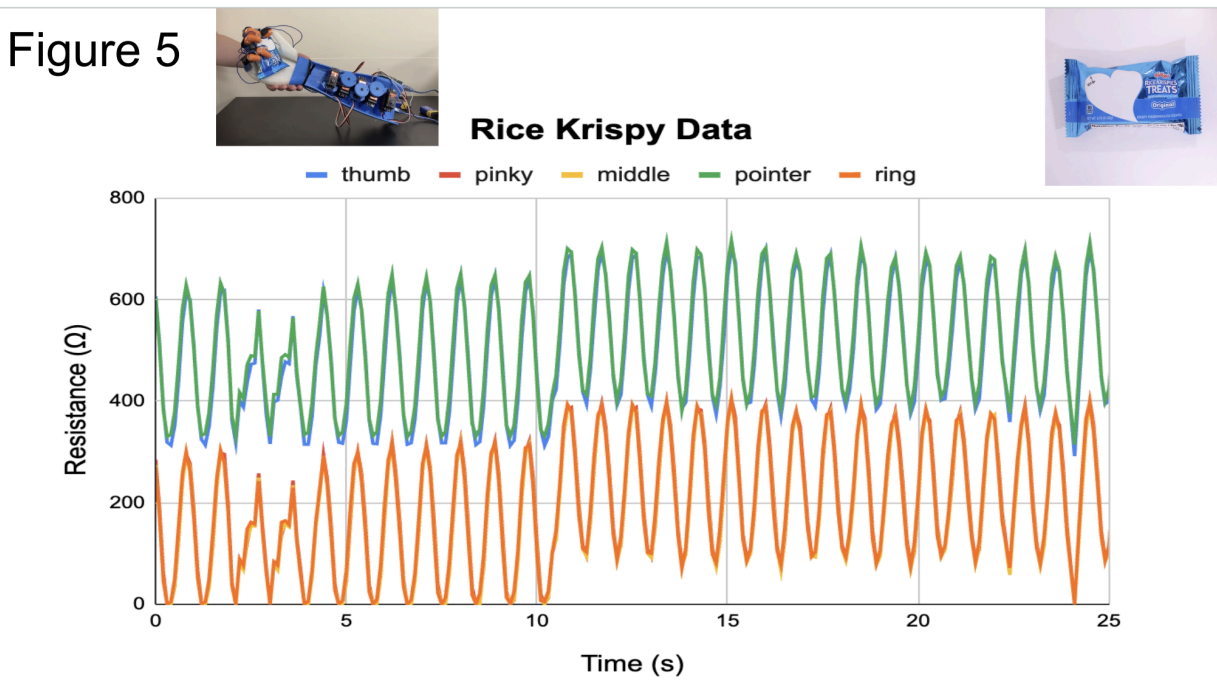


Figure 5: The Rice Krispy data, with the resistance values (in Ω) graphed by finger per second when grabbing a rice krispy bag.

Table 1. Figure 5 Rice Krispy Statistics:

Class	Mean (Ω)	Minimum (Ω)	Maximum (Ω)	Std. Deviation (Ω)	Coefficient of Variation
Thumb	467	290	735	122	26%
Pointer	480	305	742	121	25%
Middle	162	0	427	121	75%
Ring	166	0	428	122	73%
Pinky	165	0	432	122	74%

Figure 5 presents the resistance of the five pressure sensors during the grasping of a Rice Krispy bag over a 25-second period. The thumb and pointer fingers exhibit higher mean resistance values of 467 Ω and 480 Ω , respectively, compared to the middle, ring, and pinky fingers, which have mean resistance values of 162 Ω , 166 Ω , and 165 Ω , respectively. The pointer finger recorded the highest mean resistance at 480 Ω , while the middle finger recorded the lowest at 162 Ω , representing a difference of 318 Ω . The maximum resistance values further demonstrate differences in sensor readings across the fingers. The pointer finger reaches a peak resistance of 742 Ω and the thumb a peak resistance of 742 Ω . In comparison, the middle, ring, and pinky fingers reach maximum values of 427 Ω , 428 Ω ,

and 432 Ω , respectively. Thus, the specific grip pattern for the Rice Krispy is characterized by overall medium force, with the thumb and pointer significantly higher. The standard deviations are relatively similar across all five fingers, ranging from 121 Ω to 122 Ω . The deviations are similar, but the means for the middle, ring, and pinky fingers are much lower causing a higher coefficient of variation. This means the consistent amplitude between the fingers has a higher variation when compared to lower means. Overall, the quantitative results demonstrate the resistance patterns between the thumb and pointer fingers and the middle, ring, and pinky fingers during the grasping of the Rice Krispy bag.

Figure 6

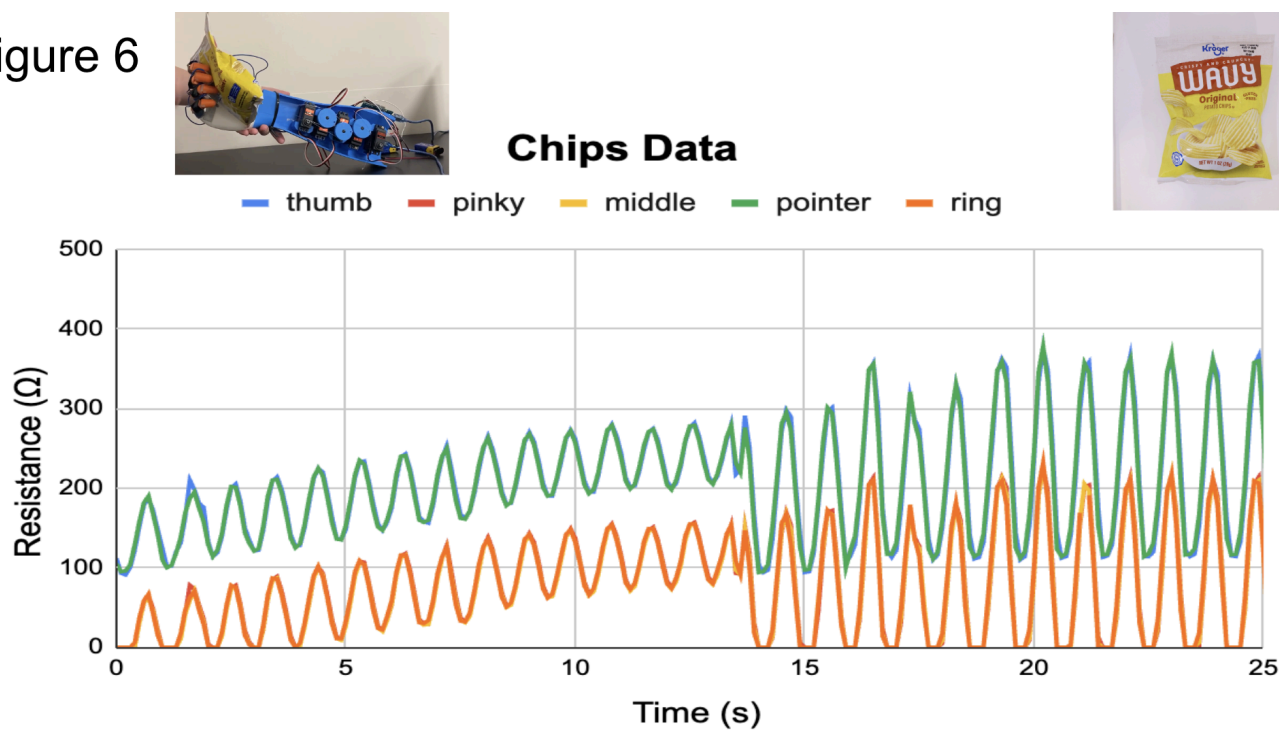


Figure 6. The chips data, with the resistance values (in Ω) graphed by finger per second when grabbing a chip bag.

Table 2. Figure 6 Chips Statistics

Class	Mean (Ω)	Minimum (Ω)	Maximum (Ω)	Std. Deviation (Ω)	Coefficient of Variation
Thumb	240	91	461	92	38%
Pointer	240	94	460	91	38%
Middle	74	0	249	74	100%
Ring	75	0	247	75	100%
Pinky	75	0	251	75	100%

Figure 6 illustrates the resistance of the five pressure sensors during the grasping of a chip bag over a 25 second interval. During the initial 14 seconds, the resistance exhibits periodic oscillations, with a gradual increase in overall resistance. After 14 seconds, the amplitude of the resistance oscillations increases dramatically, ranging from 100Ω to 400Ω on the pointer finger and 0Ω to 250Ω on the ring. This is because as the chip bag is compressed more, it becomes more difficult to compress. Additionally, the bag is comparatively malleable and the distribution of air and chips within the bag change during the grabbing process. The overall average from the thumb, pointer, middle, ring, and pinky were 240, 240, 74, 75, and 75Ω

respectively. Despite the differing trend between the first and second half, the general resistance values from gripping the chips bag are very low. The thumb and pointer contribute a stronger grip, but not much higher than the other fingers. The fingers all maintain a consistent amplitude in oscillation, causing relatively consistent standard deviations, with the thumb and pointer slightly higher. This causes a higher coefficient of variation due to the extremely low mean resistance values compared to a normal standard deviation. Generally, the grip pattern for the chips bag can be seen between the stronger thumb and pointer and weaker middle, ring, and pinky fingers.

Figure 7

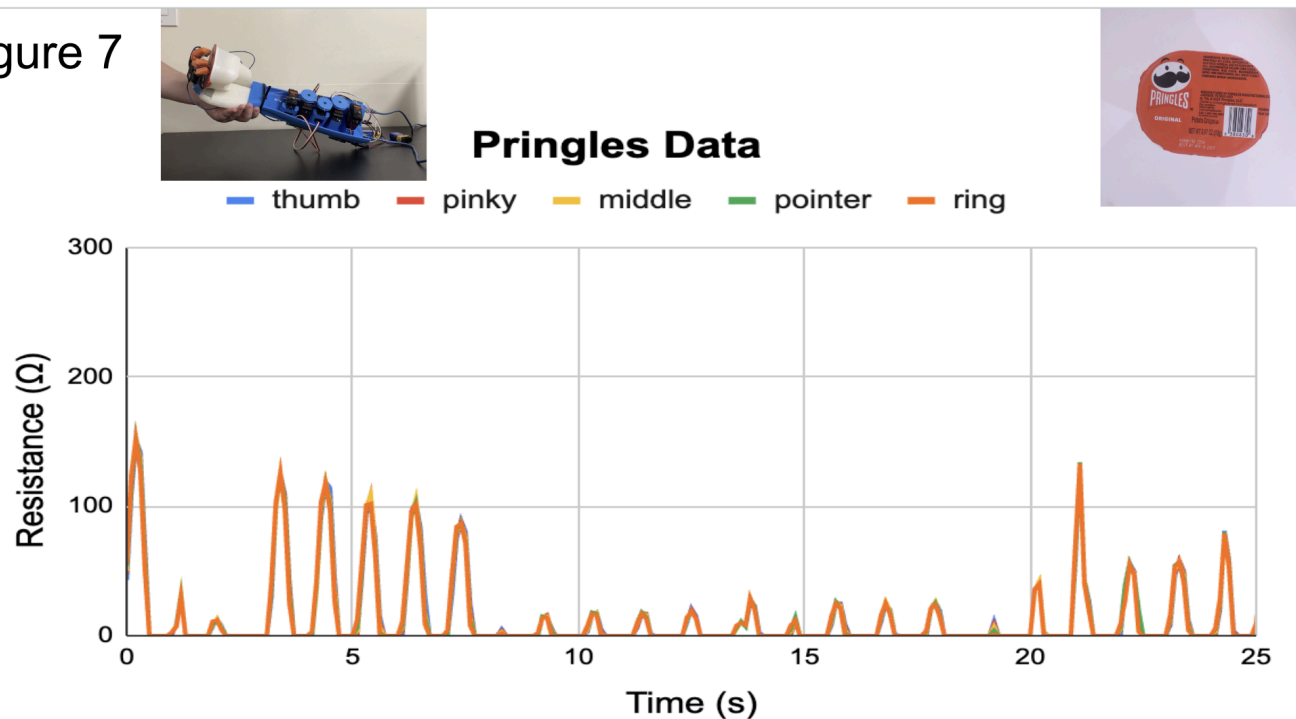


Figure 7: The Pringles Data, with the resistance values (in Ω) graphed by finger per second when grabbing a Pringle box.

Table 3. Figure 7 Pringles Data:

Class	Mean (Ω)	Minimum (Ω)	Maximum (Ω)	Std. Deviation (Ω)	Coefficient of Variation
Thumb	33	0	363	56	170%
Pointer	33	0	361	56	170%
Middle	32	0	361	55	172%
Ring	32	0	359	55	172%
Pinky	32	0	363	55	172%

Figure 7 exemplifies the unique pattern of gripping the Pringles. All the fingers have near baseline values. This is because the grip of the Pringles focuses pressure on the fingertips, so the sensors have less area touching the Pringles. The pressure data at first rapidly decreases from 150 Ω to 90 Ω then slowly lowers until 10 seconds. Then from 10 to 20 seconds all fingers maintain low pressure around 20 ohms before rising to around 150 ohm at 21 seconds then dropping again at 22 seconds before gradually increasing until 25 seconds. This is from the smooth surface of the Pringles which cause some

slippage creating more resistance. The mean of all the fingers' resistance over time is either 33 or 32, a nearly negligible amount of grip force. There is still consistent oscillation, seen from the standard deviation, which creates an extremely high coefficient of variation compared to the low mean. Overall, the fingers have consistent oscillations around a consistently low average creating the unique pattern of gripping Pringles.

Figure 8

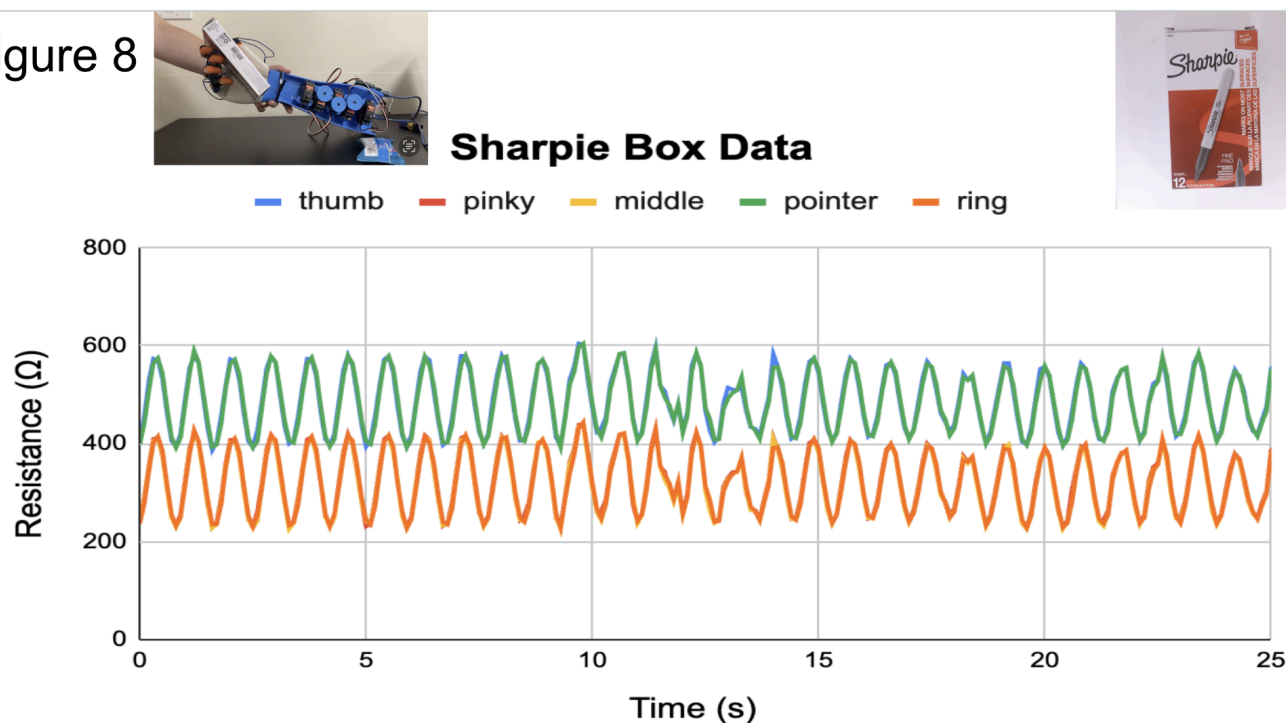


Figure 8. The Sharpie Box Data, with the resistance values (in Ω) graphed by fingers per second when grabbing a sharpie box.

Table 4. Figure 8 Sharpie Box Statistics

Class	Mean (Ω)	Minimum (Ω)	Maximum (Ω)	Std. Deviation (Ω)	Coefficient of Variation
Thumb	500	266	729	68	14%
Pointer	465	112	650	93	20%
Middle	318	107	475	64	20%
Ring	320	111	474	65	20%
Pinky	321	109	480	64	20%

Figure 8 shows the data remained consistent throughout the gripping with consistent oscillations. There was a slight change from 10 seconds to 15 seconds, showing the Sharpie box moved slightly. The averages throughout the gripping period for the thumb, pointer, middle, ring, and pinky fingers were 500, 465, 318, 320, and 321 Ω respectively. This suggests that the fingers all have a strong grip on the box, with the pointer and thumb having a tighter grip. The means are high because the box is rigid and has higher resistance against the grip. The oscillations have a great amplification, though compared to the mean creates an average standard deviation. The pointer has an unusually high standard deviation, possibly due to its position in the grip. This grip had a smaller standard deviation with all the fingers having high means leading to low coefficients of variation. The pattern for the Sharpie box is thus characterized by high resistance values with the thumb and pointer contributing more.

The Rice Krispy grip pattern is characterized by overall medium force, with the highest contributions from the thumb and pointer finger. The Chips Bag grip pattern is similar with a stronger thumb and pointer finger, but generally lower resistance values. Conversely, the Pringles grip pattern is consistently low among all the fingers yet has a high standard deviation like the other objects. The Sharpie box grip pattern is consistently high, with the thumb and pointer fingers slightly higher. Between the values for distinct fingers and the overall averages for the whole hand, a distinct grip pattern is created for every object. The random forest machine learning model takes these unique patterns and extensive data to identify whether an object is a Rice Krispy, Chips bag, Pringles, or Sharpie box.

Table 5. Accuracy of Random Forest Model in Object Detection

Training Rows: 4546; Testing Rows: 1137; Overall Accuracy: 98.1%

Class	Precision	Recall	F1-score	Support
Box	0.98	0.95	0.96	278
Chips	0.97	0.99	0.98	280
Pringles	1.00	0.99	0.99	299
Rice Krispy	0.98	0.99	0.98	280
Accuracy			0.98	1137
Macro Avg	0.98	0.98	0.98	1137
Weighted Avg	0.98	0.98	0.98	1137

Table 5 shows the accuracy of the random forest model, at 98.1%, as well the accuracy of the model for each individual object. The F1-score uses a harmonic mean to consider both the precision of the model, examining how many of the model's guesses were correct, and its recall, examining how many of the correct options were guessed by the model. The support column states the number of occurrences of each object within the dataset. The Pringles are the most accurate object with a F1-score of 0.99, because of the unique grip where the sensors barely touch the container. The other objects are also very accurate because they have unique grip patterns. Overall, the machine learning model is extremely accurate in object detection between the four objects.

This anthropomorphic hand expands on current designs by combining the foundation of a low-cost 3D printed cable actuated hand with low-cost pressure sensors. Together, these components allow for the development of pressure sensing hands that are suitable for object detection with low manufacturing costs. The total cost for this project was around \$103.07, making the hand particularly viable for the research, prototyping, and distribution of new technologies. The hand provides a highly accessible base for models like slip detection and adaptive grasp to be implemented into usable prosthetics and robotics that rely on low latency tactile sensing information.

The development of an object detection machine learning model demonstrates possible uses of the robotic hand.

Its accuracy supports the success of the hand from structure to movement to data collection. The random forest machine learning model trained on the four objects identifies the object by recognizing unique grip patterns, expanding the ability of this design to applications that require the manipulation of varying objects.

Future applications that rely on this base include adaptive intelligence in both robotic hands' grasp that can be used to support advanced prosthetics, agriculture automation for picking vegetables, cooking robots in preparing food, manufacturing tasks in an assembly line, and laboratory handling of fragile glass objects. This anthropomorphic hand can also be used in aiding low-cost computer vision, as traditional computer vision methods like Lidar fail to account for the texture and compressibility of certain materials and geometries, while pressure based tactile sensors are able to account for such variables [22,23,24].

CONCLUSION

This hand exemplifies the applicability of low-cost tactile sensing in a 3D printed anthropomorphic robotic hand. Combined with a random forest object detection model, the tactile sensors were able to consistently detect individual objects from four base cases with accuracy ranging from 97% to 100%. The model had an overall accuracy of 98.1%, supporting the viability of this technology as a future aid to adaptive grasp and computer vision assistance, especially due to its cost-effective nature compared to existing anthropomorphic tactile sensing systems.

There were many limitations in this project. The data collection was from limited grasping trials and unique objects, creating a model with limited detection capabilities. The absence of calibrated force measurements from resistance values prevents detailed analysis of data trends. Sensor variability generalized the model on possible inconsistent trends and made repeated experimentation difficult. Further data collection with more variety in grasp and objects would validate the model with improved predictive performance.

Future research should expand upon the application of tactile sensing data for active technical improvements such as adaptive grasp or computer vision. Additionally, viability of the suggested bioinspired electronic-skin finger ridge system should be explored, which could enhance the accuracy and stability of sensors in an improved grip [13,12].

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CONFLICTS OF INTEREST

[Conflict-of-interest statement not provided in the original manuscript; authors should complete this statement before submission.]

DATA AVAILABILITY

[Data-availability statement not provided in the original manuscript; authors should complete this statement before submission.]

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